

COMPANION SLIDES

Chapter 10

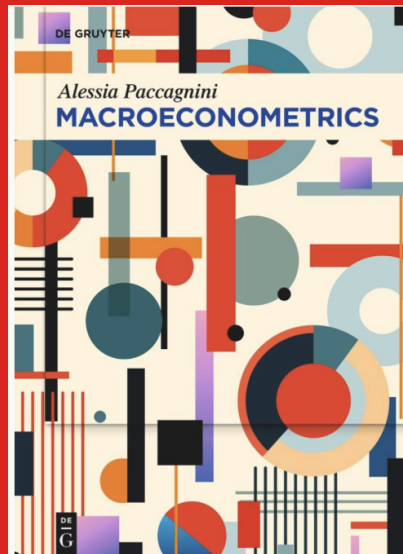
Time-Varying and Nonlinear Models

Regimes, state dependence and evolving
dynamics

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Beyond linear, constant-parameter dynamics

What if macroeconomic relationships change over time, across regimes, or with the state of the economy?

- ▶ Linear models impose a single propagation mechanism.
- ▶ Crises, policy changes and structural transformations can alter coefficients and volatility.
- ▶ Nonlinear models allow responses to depend on **time, regimes and economic states**.

When is “one model for all periods” too restrictive?

Time variation

- ▶ Stylized facts and instability
- ▶ Break and nonlinearity tests
- ▶ State-space models
- ▶ Kalman filter
- ▶ TVP-VAR with stochastic volatility

Nonlinear regimes

- ▶ Markov switching
- ▶ Threshold models
- ▶ Smooth-transition VARs
- ▶ Generalized impulse responses
- ▶ Four empirical approaches

- ▶ Explain why constant-parameter linear models may fail in macroeconomic applications.
- ▶ Distinguish structural breaks from gradual parameter drift.
- ▶ Write a state-space model and explain prediction/update steps in the Kalman filter.
- ▶ Describe TVP-VARs with stochastic volatility.
- ▶ Explain Markov-switching, threshold and smooth-transition models.
- ▶ Interpret transition probabilities, thresholds and transition functions.
- ▶ Compute and interpret generalized impulse responses in nonlinear models.
- ▶ Compare alternative definitions of macroeconomic regimes.

Observed features

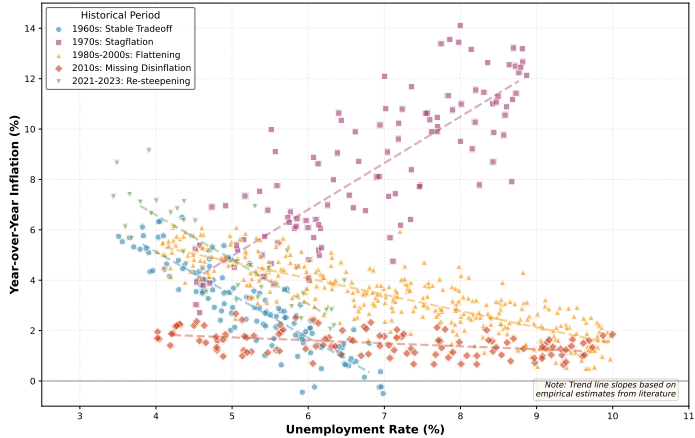
- ▶ asymmetric business cycles
- ▶ structural breaks
- ▶ evolving policy rules
- ▶ changing volatility
- ▶ crisis amplification

Modeling implication

The conditional mean, variance, or both may depend on time or on an observable/latent state.

Nonlinearity is economically meaningful when propagation itself changes.

The Evolving Phillips Curve: Parameter Instability Across Eras



Discrete change

Parameters jump at one or more break dates.
Examples: policy-regime changes,
institutional reforms.

Continuous change

Parameters evolve gradually over time.
Examples: changing transmission, slow
structural transformation.

- ▶ Break tests ask whether stability is rejected.
- ▶ Nonlinearity tests ask whether dynamics depend on states or regimes.

- ▶ Structural-break tests can target known or unknown break dates.
- ▶ Multiple-break procedures allow repeated changes in coefficients.
- ▶ Tests for neglected nonlinearity can motivate threshold or smooth-transition specifications.
- ▶ Statistical rejection is only the beginning: the nonlinear mechanism should also have an economic interpretation.

Practical principle

Use tests as diagnostics, not as a substitute for model design.

$$x_t = A_t x_{t-1} + B_t u_t, \quad y_t = C_t x_t + D_t \varepsilon_t$$

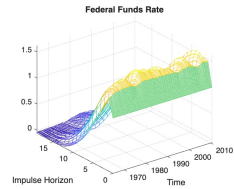
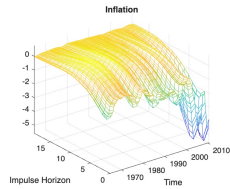
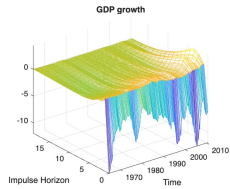
- ▶ The **state equation** describes evolution of latent states or coefficients.
- ▶ The **measurement equation** links those states to observed data.
- ▶ Time-varying coefficients can be treated as latent states and estimated recursively.

1. Predict the latent state using information through $t - 1$.
2. Predict the observable variable.
3. Compute the forecast error and its variance.
4. Form the Kalman gain.
5. Update the state using the new observation.
6. Repeat through the sample; smoothing uses future observations as well.

New data continuously revise beliefs about the hidden state.

$$y_t = c_t + A_{1,t}y_{t-1} + \cdots + A_{p,t}y_{t-p} + u_t$$

- ▶ VAR coefficients can evolve over time.
- ▶ Shock variances can evolve too: **stochastic volatility**.
- ▶ This separates changes in propagation from changes in shock size.
- ▶ Bayesian methods are especially useful because the latent state is high dimensional.



- ▶ Coefficients follow stochastic laws of motion rather than remaining fixed.
- ▶ Volatility evolves independently across shocks, allowing heteroskedasticity over time.
- ▶ Structural identification can be layered on top of the reduced-form TVP-VAR.
- ▶ Posterior simulation must recover both time-varying coefficients and latent volatilities.

Interpretation

A shock in 1980 need not have the same size or transmission mechanism as an apparently similar shock in 2005.

- ▶ Priors regularize large collections of time-varying coefficients.
- ▶ Gibbs and Metropolis steps can alternate between parameters and latent states.
- ▶ State simulation methods exploit the state-space representation.
- ▶ Posterior uncertainty matters because apparent parameter drift may be weakly identified.

$$y_t = \mu_{s_t} + \Phi_{s_t}(L)y_{t-1} + u_t, \quad s_t \in \{1, \dots, K\}$$

- ▶ Regime s_t is latent.
- ▶ Each regime can have different means, dynamics and variances.
- ▶ Regime transitions are probabilistic rather than chosen deterministically from an observed variable.

$$p_{ij} = P(s_t = j \mid s_{t-1} = i)$$

- ▶ Diagonal elements measure regime persistence.
- ▶ Expected regime duration increases with the stay probability.
- ▶ Filtered probabilities use information through t ; smoothed probabilities use the full sample.
- ▶ Transition probabilities may themselves depend on covariates.

$$y_t = \begin{cases} f_1(y_{t-1}, \dots) + u_t, & z_{t-d} \leq c, \\ f_2(y_{t-1}, \dots) + u_t, & z_{t-d} > c. \end{cases}$$

- ▶ The regime is determined by an **observable transition variable** z_{t-d} .
- ▶ The threshold c divides the state space.
- ▶ Threshold models generate abrupt changes in dynamics.
- ▶ They are natural for recession/expansion or stress/normal-state applications.

$$y_t = [1 - F(z_t)]f_1(\cdot) + F(z_t)f_2(\cdot) + u_t$$

- ▶ $F(z_t) \in [0, 1]$ provides a continuous weight between regimes.
- ▶ The threshold controls the location of the transition.
- ▶ The smoothness parameter controls how abrupt the switch is.
- ▶ As smoothness becomes very large, an ST model approaches a threshold model.

- ▶ STVARs extend smooth-transition ideas to multivariate dynamics.
- ▶ VAR coefficients become weighted combinations of regime-specific coefficient matrices.
- ▶ The transition variable can encode the business cycle, uncertainty or financial stress.
- ▶ Identification and impulse responses must account for the state at the time of the shock.

Key distinction

The same structural shock can propagate differently depending on the initial state.

$$GIRF(h, \delta, \Omega_{t-1}) = E[y_{t+h} \mid \delta, \Omega_{t-1}] - E[y_{t+h} \mid \Omega_{t-1}]$$

- ▶ In nonlinear models, responses depend on the history and initial state.
- ▶ Superposition generally fails: doubling a shock need not double the response.
- ▶ Monte Carlo simulation averages over future shocks and endogenous regime paths.

Asymmetry

Positive and negative shocks can generate different responses.

Regime dependence

A shock in recession can differ from the same shock in expansion.

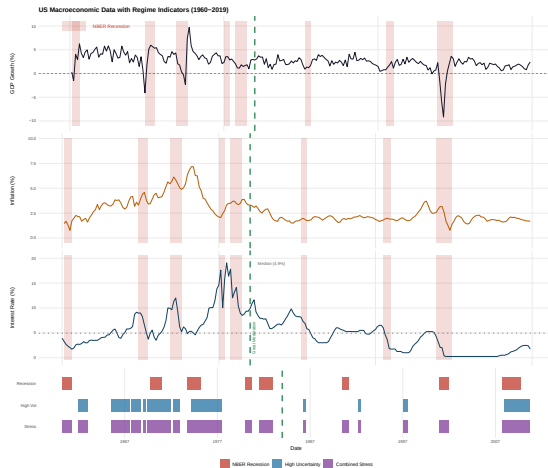
Shock-size dependence

Large shocks may move the economy into a different region of the state space.

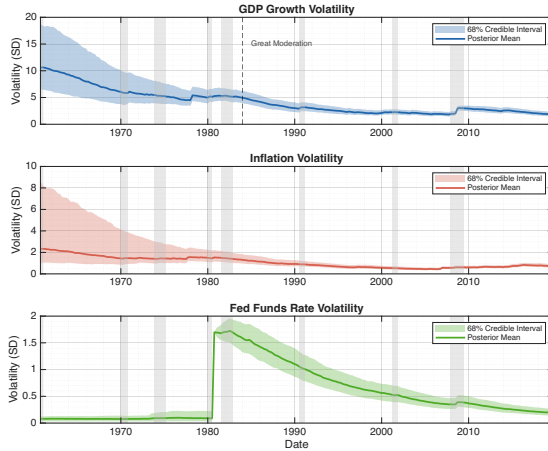
Path dependence

Future regimes can react endogenously to the shock.

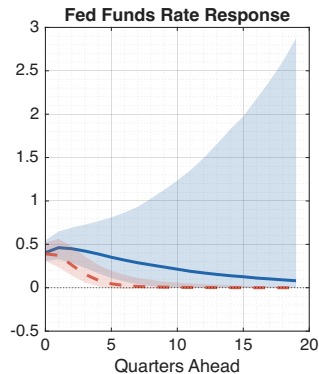
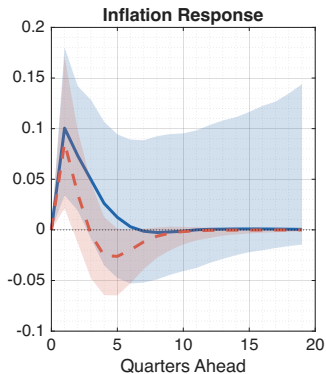
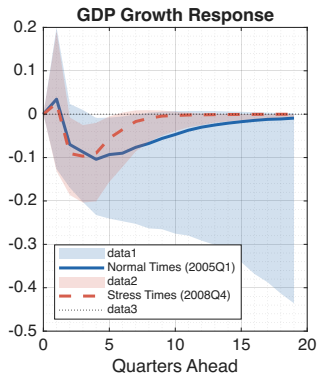
- ▶ US quarterly macroeconomic data: GDP growth, inflation and Federal Funds rate.
- ▶ Sample: 1960–2019 in the chapter application.
- ▶ Four complementary models: **TVP-VAR-SV, MS-VAR, TVAR and STVAR.**
- ▶ Objective: compare alternative ways of representing regime-dependent monetary transmission.

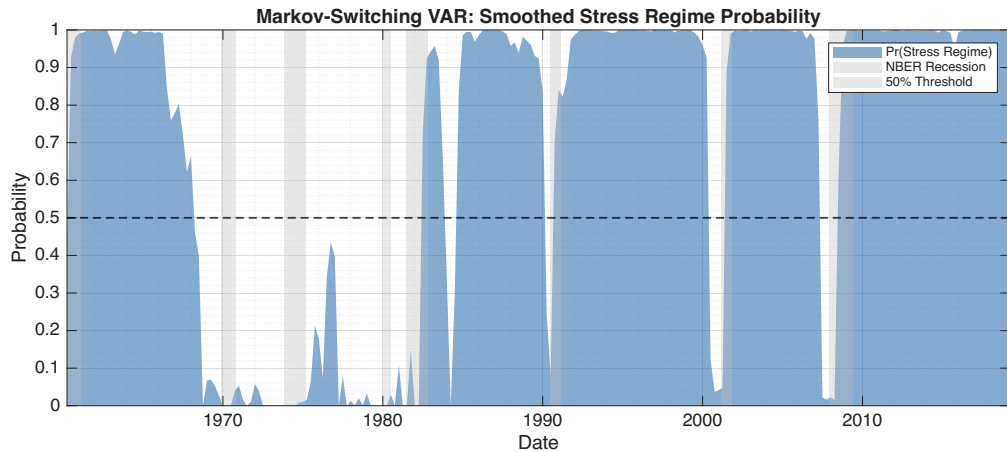


Time-Varying Stochastic Volatilities (TVP-VAR-SV)

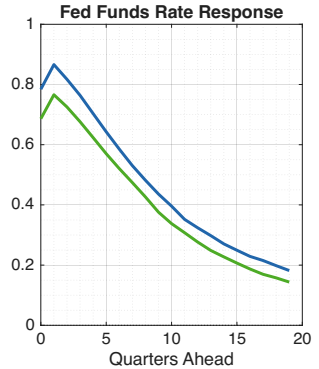
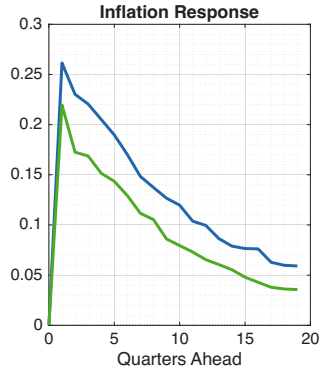
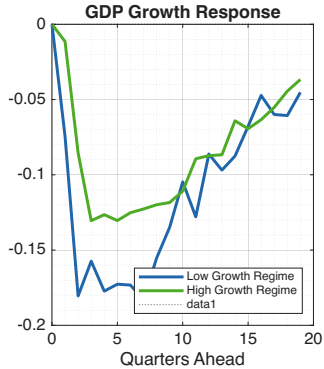


TVP-VAR-SV: Monetary Policy Shock (1 SD Tightening)
Normal vs. Stress Period Comparison



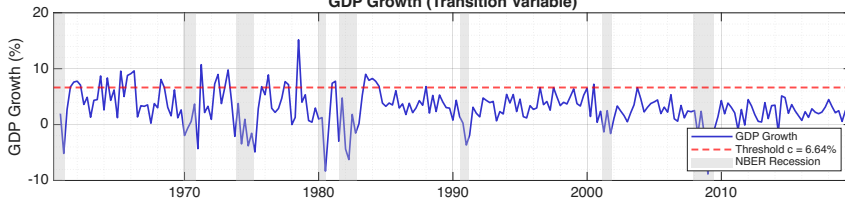


Threshold VAR: Generalized Impulse Responses
Monetary Policy Shock (1 SD Tightening)

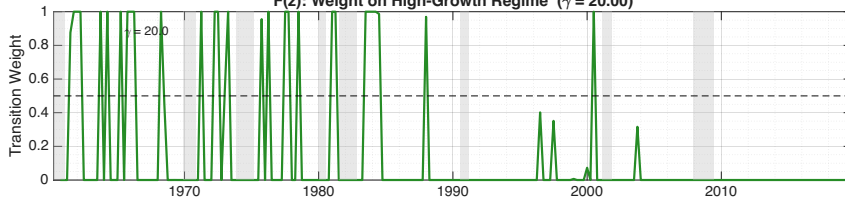


STVAR: Gradual Regime Changes

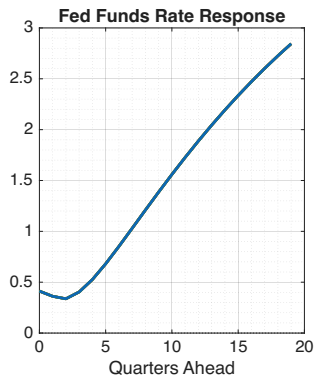
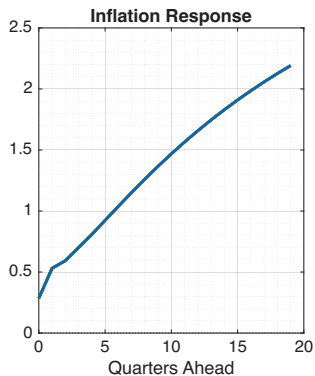
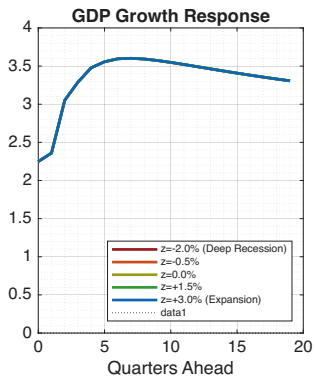
GDP Growth (Transition Variable)



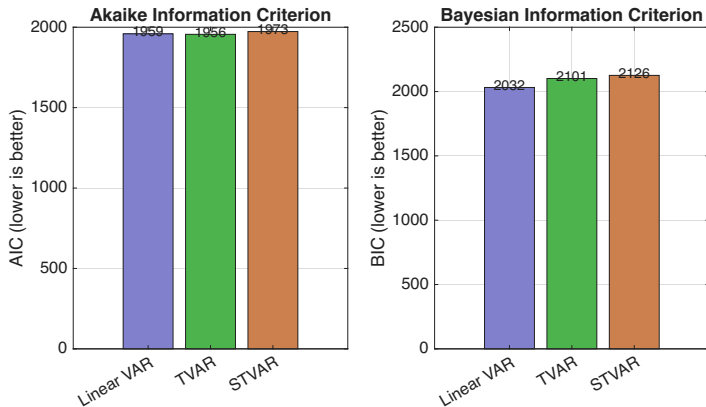
F(z): Weight on High-Growth Regime ($\gamma = 20.00$)



STVAR: Impulse Responses Across Continuum of States
Monetary Policy Shock (1 SD Tightening)

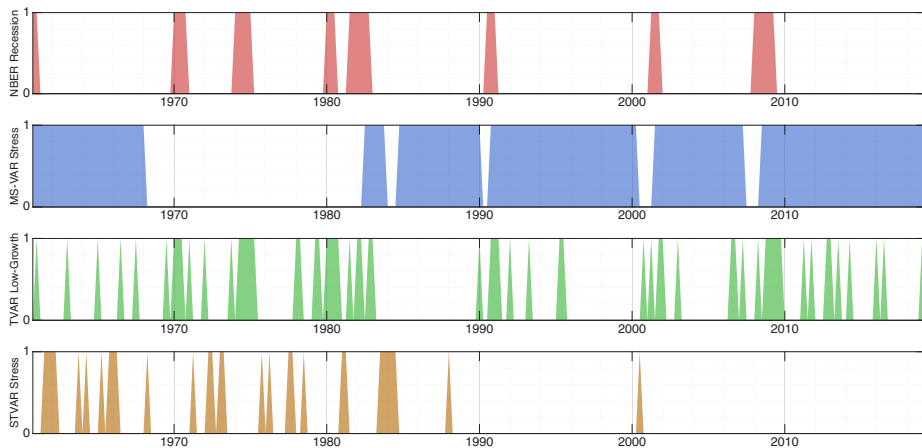


Model Selection: Information Criteria



AIC and BIC can disagree because richer nonlinear specifications improve fit while paying a larger complexity penalty.

Regime Classifications Across Four Approaches



- ▶ Time variation can appear through coefficients, volatilities, latent regimes or observed thresholds.
- ▶ Different regime definitions capture different dimensions of macroeconomic stress.
- ▶ State-dependent responses can be economically important even when model-selection criteria disagree.
- ▶ Comparing several nonlinear representations is useful because regime classification is itself part of the empirical question.

1. Start with economic reasons to expect instability or asymmetry.
2. Inspect data and test for breaks/nonlinearity.
3. Choose whether regimes are latent, observed-threshold, smooth, or continuously time varying.
4. Estimate and diagnose regime persistence/state evolution.
5. Compute state-dependent or generalized impulse responses.
6. Compare alternative models and regime classifications.
7. Stress-test results to transition variables, thresholds and identification choices.

1. Macroeconomic dynamics can change across time and states.
2. State-space methods provide a natural framework for gradual parameter evolution.
3. Markov switching models latent discrete regimes; threshold models use observed state variables.
4. Smooth-transition models bridge linear and discrete-regime dynamics.
5. In nonlinear models, impulse responses are conditional on history, state and sometimes shock size.

- ▶ Estimate a break date and compare pre/post-break VARs.
- ▶ Plot filtered and smoothed probabilities from a two-regime Markov-switching model.
- ▶ Estimate a threshold using an alternative transition variable such as uncertainty or financial stress.
- ▶ Vary the STVAR smoothness parameter and study how the transition function changes.
- ▶ Compare linear IRFs with state-dependent GIRFs.
- ▶ Re-estimate the empirical example excluding the COVID period or extending the sample.

Companion slides = starting point

These slides summarize core concepts and selected examples from Chapter 10. They are intentionally editable and are not a substitute for the full chapter.

- ▶ Add your own crisis episodes, regimes and transition variables.
- ▶ Add Python, R or MATLAB demonstrations.
- ▶ Add alternative TVP, MS-VAR, TVAR and STVAR applications.
- ▶ Add exercises on estimation, diagnostics and state-dependent impulse responses.

Make them yours.

Mixed-Frequency Models

Combining information observed at different frequencies

Next question

How can we use monthly, quarterly and higher-frequency information without throwing useful timing information away?